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 Studierichting: Informatica
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$$\text{Opgave: } \lim_{x \rightarrow 0} \frac{\sqrt[5]{1+x} - \sqrt{1+x}}{2x}$$

$$\Rightarrow y = 1 + x \xrightarrow{x \rightarrow 0} y \rightarrow 1$$

$$\Rightarrow \lim_{y \rightarrow 1} \frac{y^{\frac{1}{5}} - y^{\frac{1}{2}}}{2(y-1)}$$

$$\Rightarrow z = y^{\frac{1}{10}} \xrightarrow{y \rightarrow 1} z \rightarrow 1$$

$$\Rightarrow \lim_{z \rightarrow 1} \frac{z^2 - z^5}{2(z^{10} - 1)} = \lim_{z \rightarrow 1} \frac{z^2(1 - z^3)}{2((z^5)^2 - 1^2)} = \lim_{z \rightarrow 1} \frac{z^2(1 - z^3)}{2(z^5 - 1)(z^5 + 1)}$$

$$1 - z^3 \Rightarrow$$

$$\begin{array}{c|cccc} & -1 & 0 & 0 & 1 \\ 1 & \downarrow & -1 & -1 & -1 \\ \hline & -1 & -1 & -1 & 0 \end{array}$$

$$z^5 - 1 \Rightarrow$$

$$\begin{array}{c|cccccc} & 1 & 0 & 0 & 0 & 0 & -1 \\ 1 & \downarrow & 1 & 1 & 1 & 1 & 1 \\ \hline & 1 & 1 & 1 & 1 & 1 & 0 \end{array}$$

$$\Rightarrow \lim_{z \rightarrow 1} \frac{z^2(z-1)(-z^2-z-1)}{2(z-1)(z^4+z^3+z^2+z+1)(z^5+1)}$$

$$= \lim_{z \rightarrow 1} \frac{z^2(-z^2-z-1)}{2(z^4+z^3+z^2+z+1)(z^5+1)} = \frac{-3}{20}$$

References